Machine Learning Lab Algorithms

Short and easy algorithms — Experiments 1 to 11

Experiment 1: Basic Feature Engineering

1. Start.
2. Load the Titanic dataset.
3. Identify and handle missing values.
4. Encode categorical data into numerical values.
5. Scale numerical features.
6. Display the cleaned dataset.
7. Stop.

Experiment 2: PCA and LDA

1. Start.
2. Load the Wine dataset.
3. Separate features and target labels.
4. Split the data into training and testing sets.
5. Standardize the features.
6. Apply PCA and LDA to reduce dimensions.
7. Train a classifier and compare accuracy and training time.
8. Stop.

Experiment 3: Multiple Linear Regression

1. Start.
2. Load the California Housing dataset.
3. Separate input features and target values.
4. Split the data into training and testing sets.
5. Train a Linear Regression model.
6. Predict the target values.
7. Calculate MSE, RMSE, and R^2 score.
8. Stop.

Experiment 4: Polynomial Regression

1. Start.
2. Load the Auto MPG dataset and remove missing values.
3. Select horsepower as input and MPG as target.
4. Split the data into training and testing sets.
5. Create polynomial features of different degrees.
6. Train and test a regression model for each degree.
7. Compare MSE and select the best degree.
8. Stop.

Experiment 5: Ridge and Lasso Regression

1. Start.
2. Load the Diabetes dataset.
3. Separate features and target values.
4. Split the data and standardize the features.
5. Train the OLS regression model.
6. Train Ridge and Lasso models using different alpha values.
7. Compare MSE, R^2, and model coefficients.
8. Stop.

Experiment 6: Logistic Regression

1. Start.
2. Load the Breast Cancer dataset.
3. Separate features and target labels.
4. Split the data into training and testing sets.
5. Standardize the features.
6. Train the Logistic Regression model.
7. Predict the class labels.
8. Calculate the confusion matrix, accuracy, precision, recall, and F1-score.
9. Stop.

Experiment 7: Support Vector Machine (SVM) Classifiers

1. Start.
2. Load the dataset and separate features X and target labels y.
3. Split the data into 80% training and 20% testing sets using stratified sampling.
4. Standardize the features using StandardScaler.
5. Train SVM classifiers using Linear, RBF, and Polynomial kernels.
6. Record the support vectors and predict labels for the test data.
7. Calculate accuracy, precision, recall, F1-score, and ROC-AUC for each kernel.
8. Compare the results in a table and visualize the decision boundaries.
9. Stop.

Experiment 8: K-Means Clustering

1. Start.
2. Load the Mall Customer dataset.
3. Select Annual Income and Spending Score.
4. Standardize the features.
5. Apply K-Means for different values of k, from 2 to 10.
6. Calculate WCSS and Silhouette Score for each k.
7. Select the optimal number of clusters.
8. Display the clusters and centroids.
9. Stop.

Experiment 9: DBSCAN Clustering

1. Start.
2. Load and standardize the dataset.
3. Calculate nearest-neighbor distances and plot the k-distance graph.
4. Select suitable values of epsilon and MinPts.
5. Apply the DBSCAN algorithm.
6. Identify clusters and noise points.
7. Visualize and analyze the clusters.
8. Stop.

Experiment 10: Random Forest and Boosting

1. Start.
2. Load the Breast Cancer dataset.
3. Separate features and target labels.
4. Split the data into training and testing sets.
5. Train Logistic Regression, Random Forest, and Gradient Boosting models.
6. Predict the test data.
7. Compare accuracy, precision, recall, F1-score, and training time.
8. Stop.

Experiment 11: Q-Learning

1. Start.
2. Create a 4 × 4 Grid World with start and goal states.
3. Initialize the Q-table with zeros.
4. Set learning rate, discount factor, and exploration rate.
5. Select actions using the ε-greedy strategy.
6. Observe the reward and next state.
7. Update the Q-table using the Q-Learning equation.
8. Repeat for multiple episodes until training is complete.
9. Find the optimal policy and compare rewards and success rates.
10. Stop.